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Does Evolutionary Biology Need a Rethink in the Genomic Era?

The foundations of evolutionary biology were established during the Modern Synthesis, which unified Darwinian natural selection with Mendelian genetics into a coherent framework. In this view, evolutionary change results from the interaction of mutation, natural selection, genetic drift, and gene flow, with adaptation proceeding primarily through the substitution of beneficial alleles shaped by selection. Across decades, this framework successfully explained patterns in morphology, phylogeny, and quantitative traits, establishing itself as the dominant paradigm in evolutionary thinking. Yet the past twenty years have brought a revolution in genomic data, ancient DNA, and computational models. Patterns such as pervasive introgression, extensive standing variation, and the overwhelming influence of nonadaptive processes have called into question whether the Modern Synthesis fully captures the mechanisms driving biological diversity today. This raises an important question: does evolutionary biology, as traditionally formulated, require a rethink? In this essay, I argue that it does—not because classical principles are wrong, but because genome-scale data reveal that evolution operates through processes that the Modern Synthesis did not fully anticipate. By synthesizing population-genetic theory, empirical case studies, and comparative genomic datasets, I show that although classical models remain powerful, they are incomplete and require substantial expansion to accommodate the complexity uncovered in the genomic era.

Early population-genetic theory generated clear expectations for how evolutionary change should occur. Fisher's fundamental theorem suggested that natural selection predictably increases mean fitness in proportion to additive genetic variance (Fisher 1930). Haldane's formulation of fixation probability predicted that even weakly beneficial new mutations have an appreciable chance of fixing in a population, with a probability of roughly 2s under reasonable assumptions (Haldane 1927). These frameworks, along with Wright's shifting-balance theory emphasizing the interplay of drift and selection on rugged fitness landscapes (Wright 1932), form the conceptual backbone of the Modern Synthesis. In these classical models, adaptive evolution typically occurs through the sequential substitution of beneficial mutations, while gene flow is viewed largely as a homogenizing constraint on local adaptation.

However, theoretical developments since the mid-20th century highlight important limitations of this classical picture. The neutral theory argued that most molecular substitutions are not adaptive, but rather fixed by drift (Kimura 1968). Empirical estimates of effective population size reveal enormous variability across species, implying that drift is the dominant force in many lineages with small Ne. Moreover, high-resolution fitness landscapes generated through deep mutational scanning show rugged genotype-to-fitness maps with abundant epistasis, contradicting the notion that populations reliably climb smooth adaptive peaks (Fowler and Fields 2014). Together, these theoretical insights indicate that evolution often proceeds through nonadaptive processes and through exploiting standing or introgressed variation rather than repeated sweeps of new beneficial alleles. These observations challenge the completeness of classical theory and motivate reevaluating whether the Modern Synthesis adequately explains the dynamics seen in genomic datasets.

The most compelling arguments for rethinking evolutionary biology come from empirical studies where genome-scale data reveal processes that classical theory did not

anticipate. One of the strongest cases is the origin of the high-altitude EPAS1 allele in Tibetans. Huerta-Sánchez et al. (2014) demonstrated that the Tibetan haplotype is nearly identical to a Denisovan-like sequence and exhibits extreme allele-frequency divergence from Han Chinese populations, with a pronounced FST peak across a 32.7 kb region. This indicates that high-altitude adaptation in Tibetans arose not from a new mutation, but from strong positive selection acting on introgressed archaic variation. Adaptive introgression is now documented across a range of systems, including humans, canids, and Heliconius butterflies, yet it was not considered a typical or central mechanism in the Modern Synthesis. The EPAS1 example illustrates how classical expectations of adaptation via de novo beneficial mutations are insufficient to explain empirically observed evolutionary trajectories.

Genome-wide patterns of human variation further challenge classical assumptions. Lewontin's seminal analysis showed that less than 15% of human genetic variation occurs between populations, with the vast majority found within populations (Lewontin 1972). More recent whole-genome studies confirm these patterns, demonstrating low FST across continental groups despite dramatic environmental differences (Rosenberg et al. 2002). These findings imply that drift, demography, and migration—not extensive local adaptation—shape most genomic differences among human populations. Such results contradict early expectations that geographically separated groups should exhibit strong adaptive divergence. Instead, they highlight the central role of neutral and historical processes in shaping genetic diversity, phenomena that require more explicit integration into evolutionary models.

Parallel evolution in sticklebacks offers a third line of evidence supporting the need for a theoretical expansion. Repeated loss of armor plates in freshwater environments occurs through selection on standing genetic variation, particularly alleles at the Ectodysplasin locus (Colosimo et al. 2005). This suggests that evolution often proceeds by sorting and reusing pre-existing alleles rather than generating novel mutations. Although compatible with

classical theory, this mode of adaptation was underappreciated in the original Modern Synthesis, which focused on new mutations as the primary substrate of adaptation. The predictability and speed of stickleback evolution indicate that models emphasizing standing variation and its interaction with environmental change are essential for understanding real-world adaptation.

Despite compelling reasons to expand the Modern Synthesis, strong evidence shows that classical evolutionary theory continues to explain core evolutionary processes. Dettman et al. (2007) experimentally evolved yeast populations in divergent environments and demonstrated that ecological divergence can rapidly generate reproductive isolation, consistent with classical predictions of ecological speciation. Similarly, the Darwin's finches of the Galápagos provide clear, repeated examples of strong natural selection operating on beak morphology during environmental fluctuations (Grant and Grant 2002). These studies confirm that directional selection remains a potent force capable of driving rapid phenotypic and genetic change.

Such examples underscore that classical models capture essential mechanisms of evolutionary change. They show that a "rethink" of evolutionary theory is not needed because classical ideas are wrong, but because they are incomplete. Strong selection, mutation, and drift remain fundamental; the issue is that they do not fully account for many complexities revealed in the genomic age.

The genomic era has revealed widespread patterns that directly contradict assumptions embedded in 20th-century models. Phylogenomic analyses repeatedly show extensive introgression among species traditionally considered reproductively isolated. Heliconius butterflies, for instance, exchange mimicry-related loci across species boundaries (The Heliconius Genome Consortium 2012). Canid genomes show complex histories involving multiple introgression events between wolves, coyotes, and domestic dogs

(vonHoldt et al. 2016). Even Darwin's finches, once viewed as a near-perfect example of an adaptive radiation with clean species boundaries, exhibit pervasive hybridization (Lamichhaney et al. 2015). These findings contradict the classical expectation that species are cohesive, reproductively isolated lineages and instead show that gene flow can persist long after divergence.

Moreover, genome-wide analyses reveal that incomplete lineage sorting is common, causing discordance between gene trees and species trees. This complicates inference of evolutionary relationships and illustrates that evolution proceeds through networks of ancestral polymorphism rather than simple bifurcating trees. Linked selection and background selection further reduce diversity across large genomic regions (Charlesworth et al. 1993), reshaping patterns of polymorphism in ways that classical models never explicitly incorporated.

These observations suggest that evolution must be conceptualized as a multi-scale process involving interacting forces operating across entire genomes, not just at single loci. Classical models remain necessary, but they are insufficient to explain the mosaic and reticulate patterns of variation now documented across a wide range of taxa.

Given the evidence above, evolutionary biology requires a theoretical expansion rather than a replacement of its foundational principles. Future frameworks must integrate polygenic adaptation, gene regulatory network evolution, and demographic complexity. They must also incorporate genome-wide effects such as linked selection, fluctuating selection, and the interaction between selection and standing or introgressed variation.
Machine-learning–based reconstructions of fitness landscapes, advances in pangenomics, and increasingly detailed ecological data will allow more realistic modeling of how genetic, developmental, and ecological factors interact to shape evolutionary trajectories. By

embracing these dimensions, evolutionary biology can develop predictive and mechanistic models that more accurately reflect the complexity uncovered by modern data.

In sum, the question is not whether the Modern Synthesis is wrong, but whether it is complete. Classical population-genetic models continue to explain essential features of adaptation, speciation, and drift. Yet genomic data have revealed pervasive introgression, extensive standing variation, nonadaptive processes, and complex demographic histories that cannot be fully captured by mid-20th century theory alone. These findings show that evolution is more reticulate, polygenic, and historically contingent than previously appreciated. Evolutionary biology does require a rethink—one that expands and modernizes the classical framework to integrate genomic, ecological, and developmental complexity. By doing so, the field will be better equipped to understand and predict the dynamic processes that generate and shape biological diversity.

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